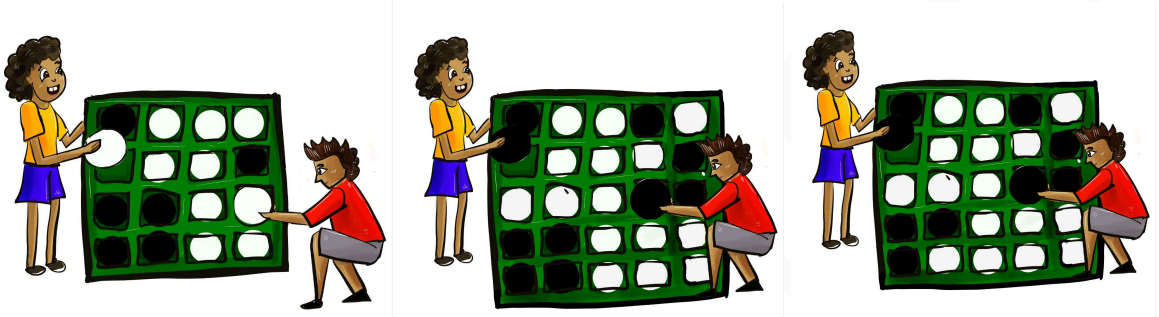
Coin Flip Magic

Grade 4: Pattern Around Us

Introduction

Using the concept of parity, which refers to the evenness or oddness of a number, we can identify the change in certain patterns. These methods are used to analyse errors in our communication systems. Here, we will explore an activity in which we play with two coloured coins (Black and White) arranged in a grid to mimic data. We will flip a coin to simulate an error in the data. The activity describes a procedure for precisely detecting errors in the grid pattern.



Activity	Time	Description
Launch	5 min	Teacher demonstrate the activity Supporting Links: Activity Video Link: https://tinyurl.com/flipmagicvideo Reference: https://tinyurl.com/fliptrickreference
Trial by Students	15 min	Student's Tryout the activity with each other. Student Worksheet: https://tinyurl.com/flipmagicworksheet
Discussions and Explorations	15 min	Attempting the worksheets and the discussion based on the activity

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CT Components

Algorithmic Thinking:
When we count the number of black coins in each row, one by one, and check whether the number is even or odd, we do the same thing for every column. Repeating these steps for every row and column follows a clear algorithm to pinpoint the flipped coin.

Decomposition:
To identify the flipped coin, we break it down into steps: focus on one row, ask whether it has an even or odd number of black cards, and do the same for every column.

Generalisation:
Once students understand the algorithm for identifying flipped coins, they will be able to find a flipped coin on any size grid.

Logic:
By flipping any coloured coin, an even number of black coins in a row/column changes to an odd number.

Abstraction:
Grid patterns represent data, and errors in the data are represented by flipping a coin.

Activity: Coin Flip Magic

You have a grid and double-coloured coins with black and white sides. Ask someone to randomly arrange the coins, white side up or black side up, in a 4×4 grid pattern. This will ensure that you are not familiar with the pattern. You will add one more row and one more column, creating a 5×5 grid pattern with the following details in mind. While adding the extra row and column, ensure each row and column has an even number of black sides up (zero is also even). Close your eyes and have someone flip any one coin on the grid pattern. When you open your eyes, you can figure out which coin was flipped. Let us see how?

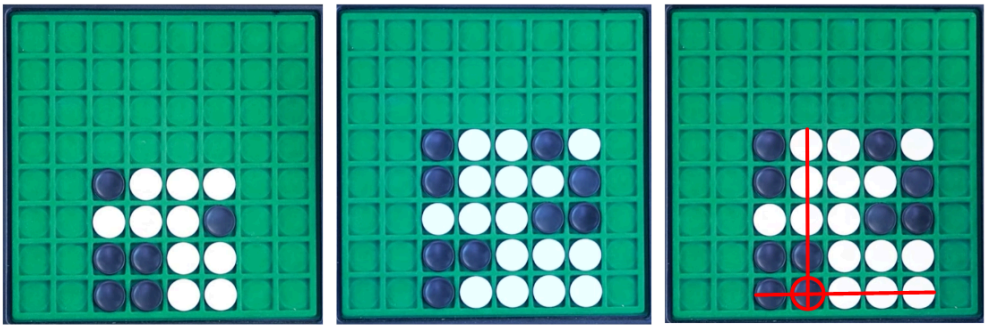


Figure: Activity Setup and Trick

When we flipped the black coin in the 5th row and 2nd column, what changes in the given row and column?

a) The number of white coins increases, and it becomes odd
b) The number of black coins decreases, and it becomes odd
c) The number of black coins increases, and it becomes odd
d) The number of white coins decreases, and it becomes odd

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Explanation

d) The number of black coins decreases, and it becomes odd

You will notice that when you flip a coin with a black side up, the number of black sides up in a given row and column decreases, and it becomes odd.

By observing rows and columns that have an odd number of black coins, we can figure out which coin was flipped in the activity.

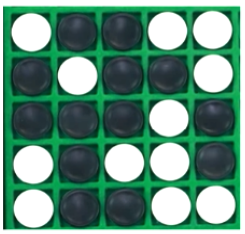
Competency :

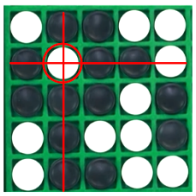
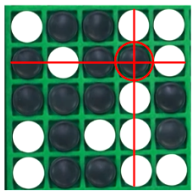
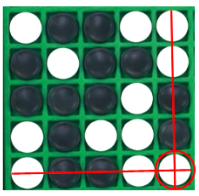
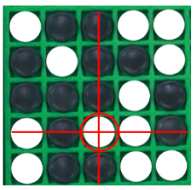
Decomposition

Logic

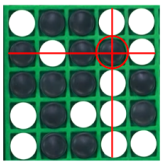
Explorations

Q. Based on what you saw above, can you identify which coin was flipped in the given grid pattern?



a)  b)  c)  d) 

Explanation

b) 

Observe that in the highlighted 2nd row and 4th column, the number of black coins is odd in number. By observing rows and columns that have an odd number of black coins, we can figure out which coin was flipped in the activity.

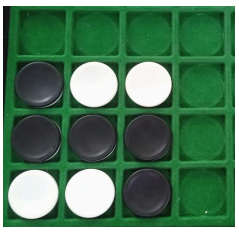
Competency :

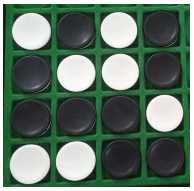
Decomposition

Logic

Q. For the given 3 x 3 grid pattern, which would be the corresponding 4 x 4 grid pattern that will help to detect which coin was flipped?

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a)  b)  c)  d) 

Explanation

c) 

In the grid pattern, observe that in all rows and columns, the number of black coins is even in number.

Competency :

Generalisation
Decomposition
Logic

Q. Instead of a 4×4 grid pattern, if we started with a 6×6 grid pattern, how many coins would be needed for the extra row and column?

a) 13
b) 14
c) 15
d) 16

Explanation

a) 13

Competency :

Generalisation
Decomposition
Logic

The Flip "trick" is really not a magic trick at all. It is a precise algorithm based on the concept of **parity**. Parity refers to the **evenness or oddness** of a number.

Remember, while adding the extra row and column, we ensured that each row and column has an even number of black coins. These extra coins are called parity coins. These parity coins help us track the “**evenness**” of the black coins before the flip and the change in “**evenness**” after the flip. This tracking is done by identifying the row and column which has an odd number of black coins.

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